

SECTION A

1 (**short**) Figure 1 shows a cylindrical tank with diameter d_1 . The tank contains an upside-down J-shaped pipe with diameter d_2 that protrudes distance h_2 below the tank and is connected to a cylinder of diameter d_3 , whose bottom end is distance h_3 from the bottom of the tank. The cylinder contains a rod and piston. The piston is a one-way valve that permits flow upwards but not downwards through the piston. The tank diameter d_1 is much greater than d_2 and d_3 .

(a) The tank, cylinder, and pipe initially contain air. The tank is filled with water until its depth, h , reaches h_1 . After filling, what are the gauge pressures at locations 1 to 5? [2]

(b) The rod is pulled up such that the J-shaped pipe completely fills with water, which starts to flow out of the protruding pipe. Stating all assumptions, what are the gauge pressures at locations 1 to 5? [2]

(c) Neglecting friction, derive an expression for the volumetric flowrate out of the tank as a function of the height, h , of water in the tank. [4]

(d) The rod is hollow. How could the rod be altered such that the tank drains to h_3 when the rod is held up but only half-drains when the rod is dropped back down as the tank drains? [2]

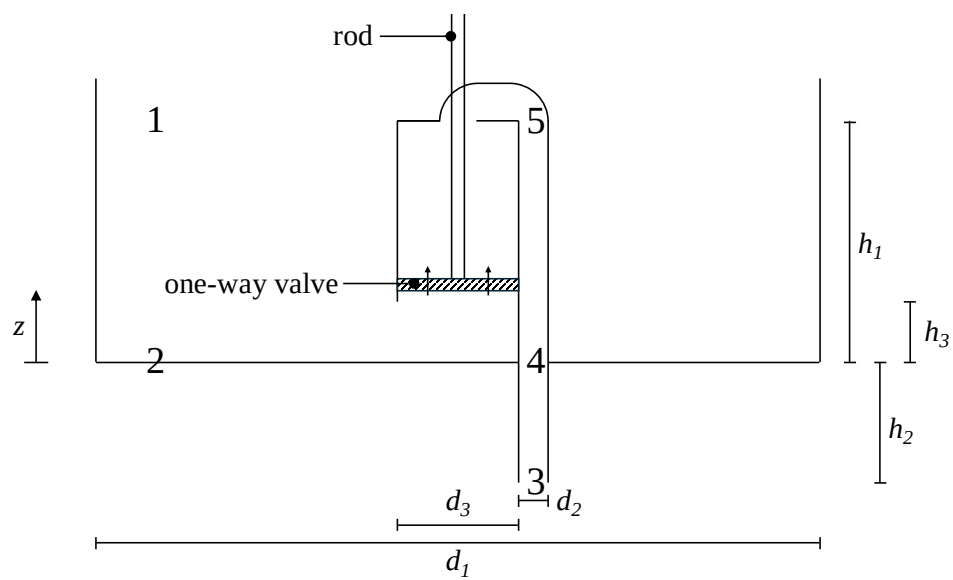


Fig. 1